BUDGETBRAIN
GLM-POWERED FINANCIAL DECISION ASSISTANT

TABLE OF CONTENTS

01 THE PROJECT

02 SYSTEM FUNCTIONALITY

03 SYSTEM ARCHITECTURE & Z AI INTEGRATION

04 ENGINEERING EXECUTION & CODE QUALITY

05 QA, TESTING & RISK MITIGATION

06 UI/UX DEMO

01

THE PROJECT

Problem Statement, Objectives and Solution

What Problem that we Observed?

Individuals and young professionals in Malaysia struggle to make informed financial decisions due to fragmented data across multiple income sources and unclear goals such as:

- Allowance
- Passive Income
- Investments

Most people track spending manually but cannot connect their daily spending to long-term goals or optimize their income sources effectively.

Targeted Domain

Domain 2: AI for Economic Empowerment & Decision Intelligence

KEY SOLUTIONS

BudgetBrain removes the friction of financial decision-making by enabling users to input their monthly spending template, monthly allowance, passive income, investments, and future goals. Using Z.AI's GLM, the system analyzes all data and provides: Recommendation, Reasoning, Expected Impact, Confidence, and a Reminder. This shifts users from "guessing about money" to "making informed decisions connected to their goals."

WHAT WE ARE WORKING ON

BudgetBrain: From Guesswork to Goal-Driven Decisions

- The Engine: Z.AI's GLM analyzes your complete financial profile (spending, income, goals).
- The Output: 5 structured insights (Recommendation, Reasoning, Impact, Confidence, Reminder).
- The Result: Replaces financial friction with clear, accountable action.

STRATEGY

BudgetBrain directly supports Domain 2 by driving economic empowerment. We go beyond basic automation to deliver true decision intelligence with quantifiable results, using our unique reminder feature to create lasting behavioral change.

MAIN GOAL OF THE SYSTEM

BudgetBrain is an AI financial partner that evaluates your:

- Spending
- income,
- investments

against your future goals. It then delivers **clear**, **actionable advice** with **built-in reminders** to keep you accountable.

TARGET AUDIENCE

YOUNG PROFESSIONALS

- Have salary + side income, want to optimize spending and investments

UNIVERSITY STUDENTS

- Have monthly allowance, want to save for goals (e.g., new laptop, travel)

ANYONE WITH PASSIVE INCOME

- Want to save money but lack financial analysis skills

GIG WORKERS

- Want to maximize returns and reinvest wisely

TARGET AUDIENCE

The Students

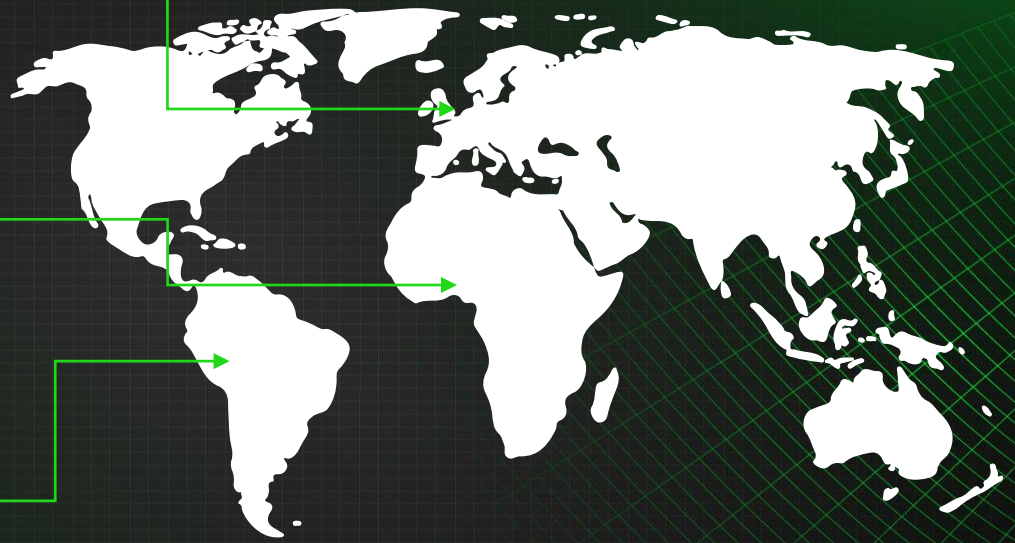
Gets an RM800 allowance but struggles to save for specific goals like a new laptop

The Young Professional

Has a salary plus side income, needing help to optimize spending and investments

The Gig Workers

Relies on fluctuating passive income and wants to reinvest it wisely



02

SYSTEM FUNCTIONALITY

The system operates as a decision intelligence engine powered by Z.AI's GLM.

The system operates as a decision intelligence engine powered by Z.AI's GLM.

5 INPUT CATEGORIES	5 OUTPUT RETUREND
TEMPLATE MONTHLY SPENDING	RECOMMENDATION – What action to take
MONTHLY ALLOWANCE	REASONING - Why this recommendation makes sense
PASSIVE INCOME	EXPECTED IMPACT - Quantifiable result (RM saved, time to goal)
INVESTMENT DETAILS	CONFIDENCE - High/Medium/Low
FUTURE GOALS	REMINDER - A specific accountability prompt

MODEL SELECTION
Why Z.AI GLM?

Multi-factor reasoning

Natural language explanation

Structured output generation

Ability to generate personalized reminders

03

SYSTEM ARCHITECTURE & Z AI INTEGRATION

The system analysis document highlights the technical scope and design-related decisions of our project.

THE KEY ELEMENT OF SYSTEM

ARCHITECTURE	BudgetBrain utilizes a simplified, database-free client-server architecture where a Streamlit frontend captures five categories of financial input and relies exclusively on Z.AI's GLM API as its only external service
DATA FLOWS	Financial inputs are packaged into a single prompt and sent to the GLM API, which returns a response that is parsed and displayed to the user as five structured outputs.
MODEL PROCESS	Users provide five financial inputs, which the Z.AI GLM instantly analyzes to generate a five-part actionable decision framework that the user can accept or reject.
ROLE OF REFERENCE	This document is a reference for developers and testers for the overall high-level architecture.

04

ENGINEERING EXECUTION & CODE QUALITY

Implementation Quality and Code Modularity

ENGINEERING EXECUTION & CODE QUALITY

CODE MODULARITY	Maintained a clean architecture by isolating all external API logic into a dedicated glm_integration.py module.
SYSTEM RESILIENCE	Built-in error handling monitors API responses. If the GLM misses the critical "Reminder" section, the system automatically retries with a lower temperature (0.1).
LATENCY AND RESOURCE MANAGEMENT	Enforces a strict 1500-token limit per session and a 10-second API timeout limit (paired with a UI loading spinner) to guarantee performance.
EMERGENCY FEEDBACK	If the Z.AI API drops below a 90% success rate during the demo, the system seamlessly falls back to a mock GLM to prevent a broken user experience.

05

QA, TESTING & RISK MITIGATION

Quality Assurance & Testing Feasibility

TESTING STRATEGY

SECTION	DESCRIPTION
Functional Testing	Validating 5 core financial inputs.
AI Output Validation	Ensuring GLM returns all 5 required sections.
Non-Functional Testing	Evaluating API timeouts, token limits, and system security.

DEPLOYMENT THRESHOLDS (PUSHING TO PRODUCTION)

CHECK	REQUIREMENT	PROJECT	PASS/FAIL
REGRESSION TEST	Minimum 90%	100%	Passed
AI OUTPUT PASS RATE	Minimum of 80% of documented prompt	100% (3/3 tests)	Passed
CRITICAL BUGS	Zero P0/01 Bugs	0 Bugs	Passed
API PERFORMANCE	Response Time < 8 second	5.2 seconds	Passed
SECURITY	API keys are not exposed	No hardcoded keys found	Passed

CHALLENGING SCENARIOS TESTED

Vague Goals

GLM successfully provides practical, structured advice.

Negative/Non-Numeric Inputs

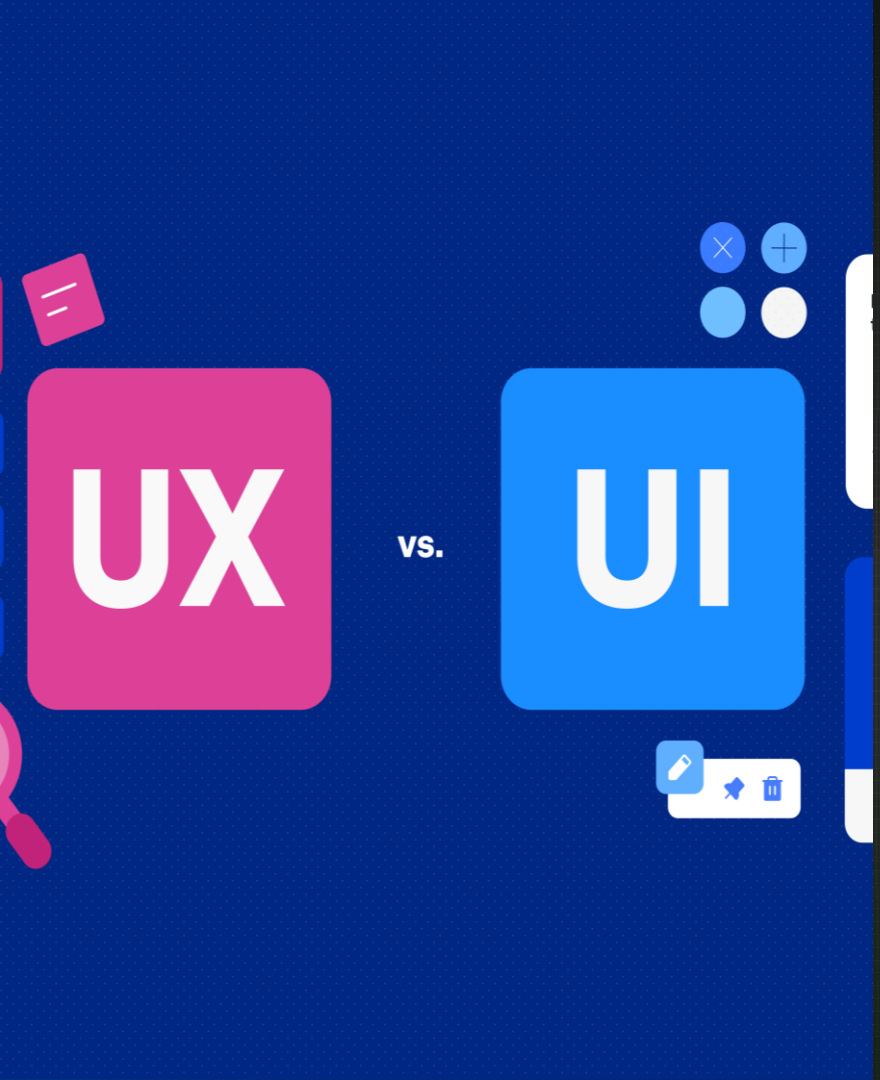
Handled gracefully; zero system crashes.

API Timeouts

Retry logic verified and working correctly.

IDENTIFIED RISKS & MITIGATIONS

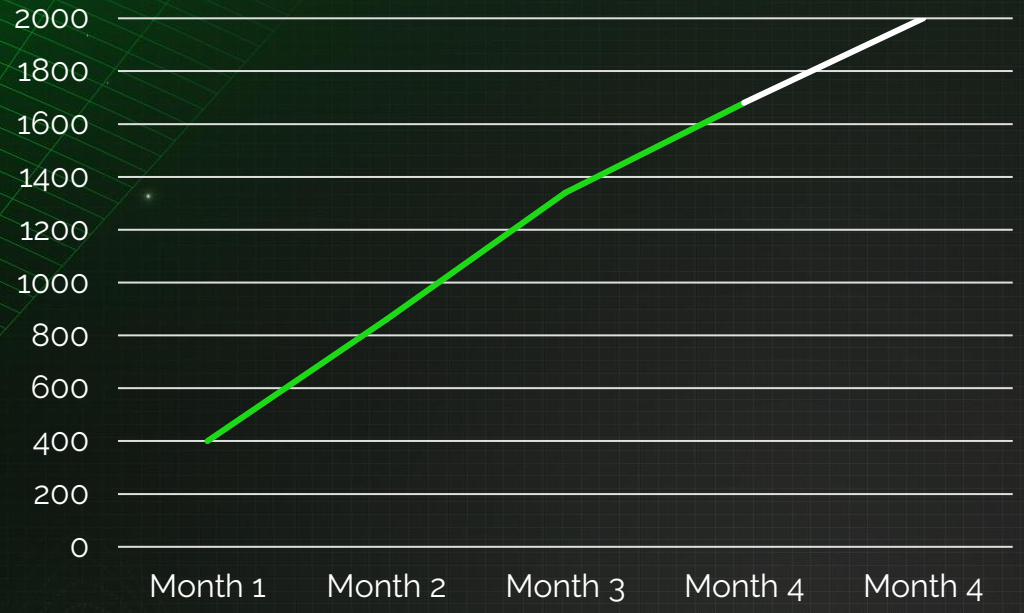
API UNAVAILABILITY	Implemented 10s timeout + retry logic.
API UNAVAILABILITY	Added auto-retry with lower temperature.
TOKEN LIMITS	System automatically summarizes spending categories.
SECURITY (API KEYS):	Keys moved to .env files; zero hardcoded keys remain.



UI/UX & DEMO

DEMONSTRATION VIDEO

PREDICTED RESULTS AFTER USING BUDGET BRAIN



Target : RM2000

Goal: Save for a new laptop in 5 months.

+RM 400 / Month

User achieves their goal precisely on time by optimizing their current fragmented income.

THE AI REMINDER

Every time you skip eating out, transfer RM20 to savings.

THANKS!

A DOMAIN 2 SOLUTION BY **TEAM FAASS TRACK**

Turning financial guesswork into goal-driven decisions